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Engineering properties of magnesium phosphate cement lunar soil concrete under vacuum conditions and its 3D printing application for lunar dome model

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ABSTRACT

To tackle the extreme environmental challenges and sustainable extraterrestrial construction material supply issues for lunar base development, this study fabricated a magnesium phosphate cement-based lunar soil concrete (L-MPC-LSC) following in-situ resource utilization (ISRU) principles and integrated it with 3D printing construction technology. This paper investigates the effects of lunar soil (LS) content on the microstructure and mechanical properties of L-MPC-LSC under both standard atmospheric and simulated lunar vacuum conditions, while evaluating the engineering feasibility of 3D-printed lunar dome model for lunar surface construction. X-ray diffraction (XRD) and scanning electron microscopy (SEM) show that LS enhances the mechanical properties of L-MPC-LSC by simultaneously modulating Dittmarite and K-struvite crystallization with providing a micro-aggregate packing effect. When the lunar-soil-to-magnesium-oxide mass ratio (LS/M) ranged from 0 to 3, Dittmarite crystals dropped from 42.43 % to zero. K-struvite crystals first increased and then decreased, peaking at 45.80 %. the changes in the microstructure lead to a nonlinear increase in the contribution rate of LS to the compressive strength of the material (67.9 %–77.3 %). Vacuum curing caused rapid dehydration, reduced K-struvite crystals, greater porosity, and 19–52 % strength loss, yet the LS/M = 3 specimen still achieved 4.1 MPa after three days—surpassing the minimum requirement for lunar-base structures and yielding 60 % ISRU. Printing trials with a lunar dome model confirmed excellent printability and the feasibility of robot-assisted construction, offering an integrated material-structure-process solution for future lunar habitats.

1. Introduction

The Moon, as the closest extraterrestrial celestial body to Earth, serves as humanity's first destination for deep space exploration.

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With major global spacefaring nations having successively launched new rounds of lunar exploration programs, the establishment of sustainable lunar bases has evolved from a scientific concept to a strategic priority in international space competition [1–7]. However, existing studies have demonstrated that the lunar surface is characterized by extreme environmental conditions, including drastic temperature fluctuations, high vacuum, microgravity, intense radiation, and frequent micrometeorite impacts [8,9]. Furthermore, the prohibitively high transportation costs between Earth and the Moon necessitate the reliance on in-situ resource utilization (ISRU) technologies for lunar base construction [10–12]. Therefore, achieving this goal critically demands breakthroughs in architectural structural design under extreme lunar environmental constraints and overcoming bottlenecks in the development of lunar regolith-based materials and intelligent construction technologies. This will drive a fundamental transformation of extraterrestrial habitation from short-term stays to long-term, sustainable living environments.

Currently, lunar base construction plans primarily revolve around three key directions: structural configurations, construction methods, and building materials. In terms of structural design, although flexible inflatable modules offer advantages such as rapid deployment and adaptability to spatial constraints, they exhibit limitations in impact resistance and long-term stability [13,14]. Although rigid module assembly can provide effective radiation shielding, it is constrained by prohibitive transportation costs and the technical challenges of on-site lunar assembly [15]. Modification approaches utilizing natural geological structures such as lunar lava tubes or impact craters, which leverage inherent environmental shielding, have emerged as a highly promising alternative. However, they still require artificial reinforcement to construct habitable spaces [16–18]. In terms of construction methodologies, while traditional large-scale manual approaches are impractical for lunar implementation, concrete 3D printing technology offers an innovative alternative. Its automation and remote-control capabilities make this intelligent construction technique an ideal candidate for extraterrestrial building projects [19,20]. Furthermore, 3D printing technology enables construction using in-situ lunar materials, significantly reducing reliance on Earth-supplied resources and aligning with sustainable exploration principles [21,22].

Despite notable advancements in the development of various lunar regolith-based concrete materials for lunar base construction, their performance remains insufficient to fully meet the stringent demands imposed by the extreme lunar environment. For instance, although polymer concrete resolves the challenge of water scarcity on the Moon, its vacuum stability and thermal cycling stability remain suboptimal [23,24]. Sulfur concrete exhibits advantages such as waterless preparation, rapid hardening, ease of fabrication, and high compressive strength. However, its thermal cycling durability demonstrates suboptimal performance under lunar conditions [25–27]. Geopolymer-based concrete exhibits promising durability under extreme lunar temperature conditions; however, its comprehensive performance under the coupled effects of multiple environmental factors requires further experimental validation [28,29]. Magnesium oxychloride cement-lunar regolith concrete exhibits reduced mechanical stability at high lunar regolith content [30,31]. Consequently, current concrete technologies still require critical breakthroughs to better address the challenges posed by the extreme lunar environment. Magnesium phosphate cement (MPC) is a cementitious material formed through acid-base and hydration reactions between magnesium oxide and phosphoric acid or water-soluble acid phosphates [32]. It exhibits advantageous properties such as rapid setting, high early-age strength, excellent dimensional compatibility, and freeze-thaw resistance [33–39]. These characteristics facilitate the mitigation of moisture evaporation in vacuum environments, rendering MPC particularly suitable for construction 3D printing under the Moon's water-scarce conditions. Furthermore, in-depth analyses of the chemical composition of LS materials have revealed that lunar soil (LS) exhibits significant similarities to fly ash in terms of chemical composition [40–45]. Studies have demonstrated that fly ash plays a beneficial role in the MPC system by effectively refining its microstructure and further enhancing the mechanical strength of MPC [38,46,47].

This study proposes a construction plan for a dome structure of a lunar lava tube base. Combining the characteristics of MPC, MPC was mixed with LS to develop magnesium phosphate cement-based lunar soil concrete (L-MPC-LSC) for 3D printing, this study aims to systematically investigate the effects of LS on the mechanical properties and underlying micro-mechanisms of L-MPC-LSC under ambient conditions and simulated vacuum environments. The ultimate goal is to determine the optimal LS content to meet the minimum strength requirements for lunar base construction, thereby maximizing the utilization of ISUR. An integrated mixing-stirring-extrusion 3D printing system [48] was independently developed, which innovatively employs dry powder feeding combined with wet material extrusion featuring instant molding. This technological advancement overcomes the limitations of conventional 3D printing pumping systems for wet materials under extreme lunar environmental conditions, enabling the successful construction of dome structure models for lunar lava tube bases. Furthermore, the printability of L-MPC-LSC materials with LS optimal dosage was systematically evaluated, demonstrating both the efficiency and applicability of this 3D printing technology for lunar base construction. Through this study, we aim to contribute to the advancement of lunar exploration and habitation research while establishing theoretical foundations and practical guidelines for future extraterrestrial construction technologies.

2. Experimental program

2.1. Raw materials

In this study, MPC was employed as the cementitious material, with its reaction system primarily consisting of light-burned magnesium oxide (MgO), industrial-grade potassium dihydrogen phosphate (KH_2PO_4), and borax retarder ($Na_2B_4O_7 \cdot 10H_2O$, denoted as B). The MgO powder, derived from magnesite calcined at 850 °C and subsequently ground, exhibited a purity of 92 % and was supplied by Dashiqiao City Jubo High-Temperature Refractory Materials Business Department. The industrial-grade KH_2PO_4, sourced from the same supplier, demonstrated 98 % purity, a particle size range of 250–350 μm after milling, and a density of 2.34 g/cm^3. The borax retarder, complying with the GB/T537-2009 standard, is a white crystalline solid with 95 % industrial purity. Ordinary tap water (W) served as the mixing water. NUAA-1A lunar soil simulant (LS), synthesized from volcanic slag obtained from Jinlongdingzi

Mountain in Huinan County, was incorporated as an admixture [49]. The chemical compositions of MgO and LS are detailed in Table 1.

To characterize the physicochemical properties of LS, this study conducted detailed analysis of its chemical composition, mineral composition, and microstructure using X-ray diffraction (XRD) and scanning electron microscopy (SEM). The sample preparation procedure was as follows: The LS powder sample was first immersed in absolute ethanol. After the initial 24-h immersion, the ethanol was replaced with fresh absolute ethanol, and the immersion continued for an additional 6 days. Subsequently, the sample was transferred to a vacuum drying oven at 40 °C for a 72-h dehydration treatment. Phase analysis was performed using an X-ray diffractometer (D/max-2500PC) equipped with a Cu Kα radiation source ($\lambda = 0.15419$ nm). The measurement parameters were set as follows: accelerating voltage of 30 kV, scanning range of 5°–70° (2θ), and a scanning rate of 5°/min. For microstructural morphology characterization, dried bulk samples (approximately 5 mm in size) were gold-sputtered and observed using a Zeiss MERLIN Compact field-emission scanning electron microscope (Germany). Fig. 1(a) reveals the XRD pattern of LS, phases of quartz, plagioclase (labradorite), kyanite, rutile, partheite, osumilite, ferrosilite, mullite and ilmenite can be observed in LS. As shown in Fig. 1(b), the LS particles derived from crushed volcanic ash exhibit irregular polyhedral morphologies. The combined presence of these mineral phases along with their characteristic microstructural signatures demonstrates that the developed simulant closely resembles Apollo-derived lunar soil samples in terms of microstructure [50].

2.2. Mix proportions

The L-MPC-LSC formulation was designed based on preliminary studies with the following key parameters: a fixed MgO/KH_2PO_4 molar ratio of 4 (M/P = 4), water-to-binder ratio of 0.24, and borax dosage equivalent to 15 % of MgO mass. To investigate the influence of LS content on the L-MPC-LSC material, LS was incorporated via the internal incorporation method (by equal volume substitution of magnesium oxide, potassium dihydrogen phosphate, and borax). The LS content ranged from 0 to 69.2 % by mass of the total L-MPC-LSC system. Within this range, the mass of LS corresponded to 0 to 4.5 times the mass of MgO. Both proportions (system mass percentage and MgO mass ratio) are provided in Table 2 for reference.

2.3. Specimen preparation and curing

The specimens were fabricated using 3D printing technology, where the extrusion process was precisely controlled to achieve desired geometries. After printing, the formed specimens were cut and polished into standardized blocks with dimensions of 40 mm × 40 mm × 160 mm to ensure uniformity and structural integrity (Fig. 2(a-b)). The specimens were then cured in two distinct environments: ambient air and a simulated vacuum, for durations of 3 h, 3 d, and 28 d prior to mechanical property testing. For vacuum curing of the 3D-printed L-MPC-LSC specimens, a vacuum-sealed packaging method was employed (Fig. 2). Upon completion of specimen preparation, the samples were immediately transferred into specialized vacuum-sealed bags. A vacuum pump was used to continuously evacuate the sealed chamber, maintaining a controlled vacuum level of -10^6 Pa throughout the curing period. Although the lunar surface exhibits an extreme vacuum of approximately 10^{-11} Pa [51], the vacuum level applied in this study (-10^6 Pa) was selected to closely approximate low-pressure environmental characteristics within feasible engineering limits.

2.4. Test methods

2.4.1. Mixing-stirring-extrusion integrated concrete 3D printing equipment

While 3D printing technology shows broad application prospects in lunar base construction, its implementation under the Moon's unique environmental conditions faces significant challenges. The extreme lunar environment-characterized by high vacuum, low gravity, and drastic temperature fluctuations-severely tests conventional extrusion-based concrete 3D printing technology. These combined environmental factors disrupt traditional pumping and rapid hardening processes, rendering them incapable of meeting the specific requirements for lunar base construction. To address these challenges, specimen fabrication was conducted using a self-developed integrated mixing-stirring-extrusion 3D printing system for rapid-setting concrete (as shown in Fig. 3(a)), engineered by Dalian Xueqing Mingfeng CNC Technology Co., Ltd. As shown in Fig. 3(b), the device innovatively adopts separate conveying pipelines for dry materials and liquid components, effectively avoiding the difficulties that traditional pumping systems might encounter in lunar environments. This design innovation enables rapid contact and mixing of dry materials and liquid within the printing head, achieving immediate material mixing, stirring, and extrusion molding, which ensures real-time material utilization and forming efficiency. Furthermore, the device's instant extrusion capability allows it to adapt to the impacts of extreme lunar surface conditions on the concrete extrusion process. By implementing the "dry powder in, wet material out" printing mode, the equipment not only solves the challenge of printing quick-setting MPC with short coagulation time, but also significantly reduces construction complexity by eliminating the need for premixed slurry preparation and minimizing water consumption. This innovative technological solution provides a groundbreaking approach for lunar base construction.

Table 1
Chemical composition (wt.%) of binders.

Oxide	MgO	SiO_2	CaO	FeO_X	Al_2O_3	TiO_2	P_2O_5	MnO	Na_2O	K_2O	Loss
MgO	85.53	6.46	1.54	0.60	0.64	–	–	–	–	–	5.23
Lunar soil	7.89	44.46	7.17	12.69	17.45	5.17	0.65	0.14	3.78	3.31	–

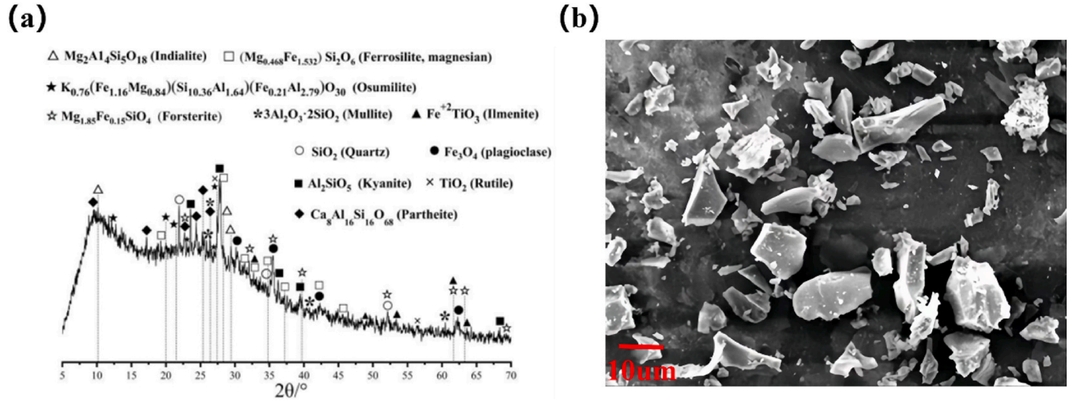

Fig. 1. (a) XRD pattern and (b) microscopic morphology of lunar soil simulant

Note: Quartz (PDF#47–1144), Plagioclase (PDF#26–1136), Kyanite (PDF#44-0027), Rutile (PDF#21–1276), Partheite (PDF#36–0378), Ferrosilite (PDF#48–0585), Indialite (PDF#12–0303),Osumilite (PDF#25–0658), Forsterite (PDF#31–0795), Mullite (PDF#15–0776), Ilmenite (PDF#29–0733).

Table 2

Mix proportion of L-MPC-LSC.

	MgO/g	KH_2PO_4/g	LS/g	B/g	LS/L-MPC-LSC	B/MgO
LS/M = 0	499.9	425.1	0	75	0 %	15 %
LS/M = 1	333.3	283.4	333.3	50	33.3 %	15 %
LS/M = 2	250	212.6	500	37.5	50 %	15 %
LS/M = 3	200	170.1	600	30	60 %	15 %
LS/M = 4	166.6	141.7	666.4	25	66.6 %	15 %
LS/M = 4.5	153.8	130.8	692.1	23.1	69.2 %	15 %

Note: LS/M denotes the mass ratio between LS and MgO.

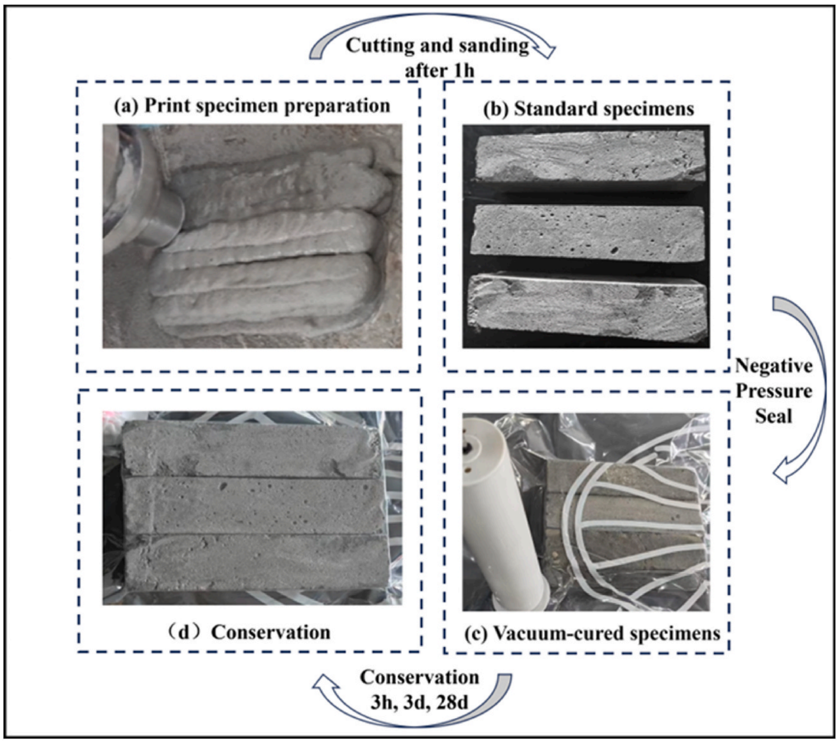

Fig. 2. Preparation of 3D Printing L-MPC-LSC specimens and vacuum environment curing process.

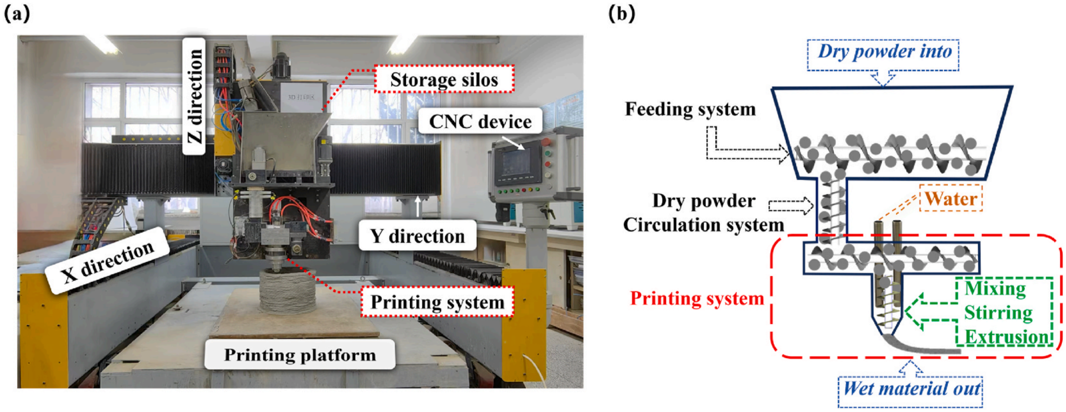

Fig. 3. (a) Mixing-Stirring-Extrusion integrated concrete rapid solidification 3D printing equipment and (b) Printing Mode.

2.4.2. *Setting time test*

In accordance with the Chinese National Standard GB/T 1346–2011 "Test Methods for Water Requirement of Normal Consistency, Setting Time, and Soundness of Cement", the setting time of the L-MPC-LSC slurry was measured using the procedures outlined in this specification. Considering the short interval between the initial and final setting times of the L-MPC-LSC slurry, this experiment selected the initial setting time as the measurement index to evaluate the slurry's setting characteristics.

2.4.3. *Mechanical performance test*

Flexural and compressive strength tests were performed in accordance with the Chinese National Standard GB/T 17671-2021 (Test

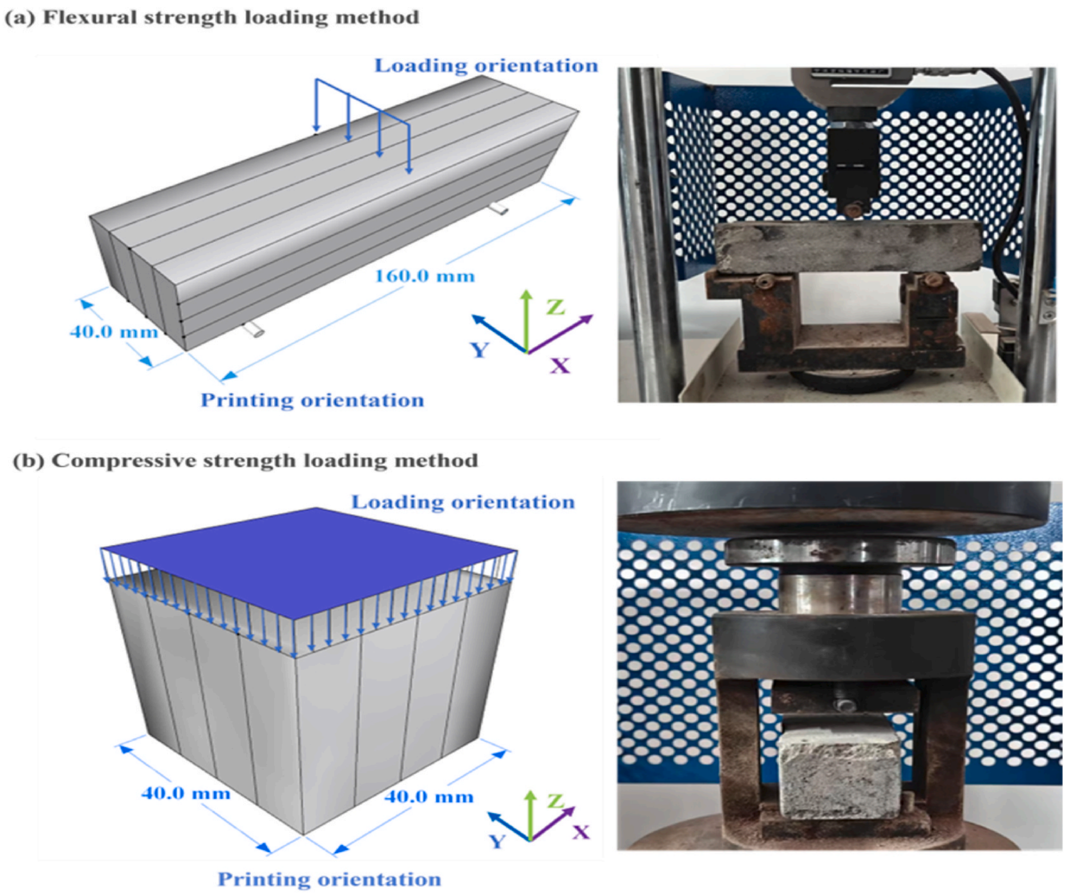

Fig. 4. Loading methods for mechanical property testing of 3D printing specimens.

methods for strength of cement mortar (ISO method)). All mechanical evaluations were conducted along the Z-axis direction, perpendicular to the printed layers, to assess anisotropic properties inherent to the 3D-printed specimens. The specific loading orientation for the 3D-printed test blocks is schematically illustrated in Fig. 4.

2.4.4. Hydration products and microstructural characterization

After mechanical testing, bulk and powdered samples were immediately immersed in anhydrous ethanol to terminate hydration. Initial immersion lasted 24 h, followed by replacement with fresh anhydrous ethanol for an additional 6 days. Subsequently, the samples were transferred to a 40 °C vacuum drying oven for 72-h dehydration. Phase analysis was conducted using an X-ray diffractometer (D/max-2500PC) equipped with a Cu/Kα radiation source (λ = 0.15419 nm). Testing parameters were set as follows: accelerating voltage of 30 kV, scanning range of 5°–70° (2θ), and scanning rate of 5°/min. Based on XRD results, quantitative analysis of phase content was performed using Topas 5.0, a full-pattern curve-fitting software developed by Bruker (Germany). For microstructural morphology characterization, dried bulk samples (approximately 5 mm in size) were gold-sputtered and observed using a Zeiss MERLIN Compact field-emission scanning electron microscope (Germany).

2.4.5. 3D printing performance evaluation

In this study, referencing the standard TCCPA 34–2022 "Test Methods for Properties of 3D Printing Concrete Mixtures," various 3D printed components were designed to evaluate the extrudability and buildability of 3D printed concrete. To assess extrudability, a printing strip (as shown in Fig. 5) was designed with dimensions of 2010 mm in length, 20 mm in width, and 10 mm in height. To assess buildability, a hollow cylindrical component as shown in Fig. 6(a) was designed, with an average diameter D of 150 mm and a height H of 204 mm. After printing, referencing the evaluation method for printability of magnesium phosphate cement (MPC) concrete proposed by Zhong [52] in integrated mixing and extrusion-based rapid 3D printing, the width and thickness of the printed strips were measured at five marked locations on Fig. 5. The average values of these five points were calculated. Furthermore, the uniformity of the printed strips was evaluated by analyzing the uniformity of their width and thickness. When the width uniformity falls within the range of 0–5 % and the thickness uniformity is within 0–1 %, it indicates good extrudability. For the measurement points shown in Fig. 6(b), the average diameter of the component's wall thickness and its height are measured, denoted as d and h, respectively. Here, d is obtained by averaging measurements from three distinct points on both the top and bottom surfaces, while h is calculated based on the average height corresponding to the three positions where the diameter is measured. The overall evaluation is then performed using the average total deviation Δk of the diameter and height. A Δk value within the range of 0 %–10 % indicates that the component's current state is suitable for printing.

The uniformity of the printed strip width and thickness is calculated using Equations (1) and (2)

$$A = \left(\frac{1}{5}\sum_{i=1}^{5}\left|\frac{K_i - a}{a}\right|\right) \times 100\% \tag{1}$$

$$B = \left(\frac{1}{5}\sum_{i=1}^{5}\left|\frac{T_i - b}{b}\right|\right) \times 100\% \tag{2}$$

In the formula, A and B denote the uniformity of the width and thickness of the printed strip, respectively, in percent (%). K_i and T_i represent the width and thickness of the printed strip at each measurement point, in millimeters (mm). a and b are the arithmetic mean values of the width and thickness of the printed strip at five measurement points, in millimeters (mm).

The dimensional deviations of the average diameter and height of the wall thickness are denoted as ΔD and ΔH, respectively, and their calculation methods are detailed in Equations (3) and (4):

$$\Delta D = \frac{|d - D|}{D} \times 100\% \tag{3}$$

$$\Delta H = \frac{|h - H|}{H} \times 100\% \tag{4}$$

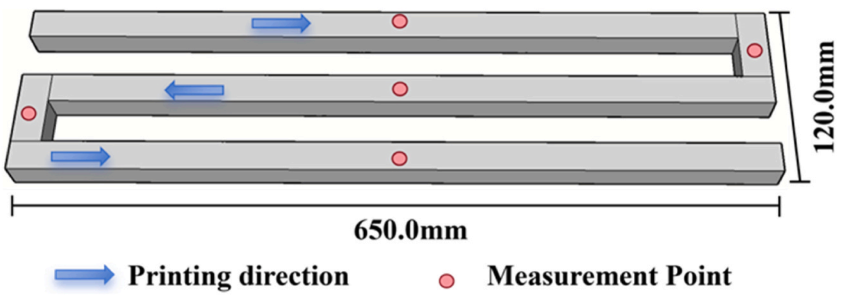

Fig. 5. 3D printing extrusion test path and measurement position distribution.

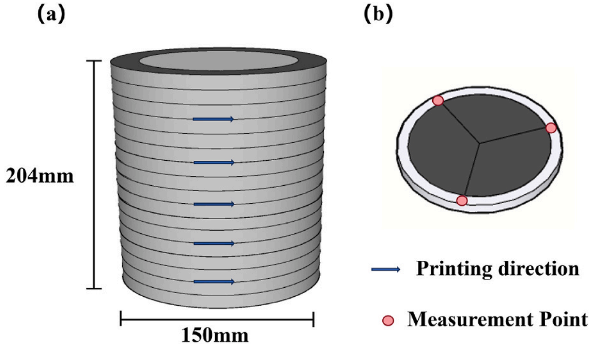

Fig. 6. (a) 3D printing construction path and (b) measurement position distribution.

The overall mean deviation of diameter and height is denoted as ΔK, and its calculation method is given in Equation (5):

$$\Delta K = \frac{\Delta D + \Delta H}{2} \tag{5}$$

3. Results and discussion

3.1. Setting time of L-MPC-LSC

Fig. 7 shows the effect of different LS dosages on the setting time of L-MPC-LSC. The results demonstrate that variations in LS dosage significantly influence the setting time of L-MPC-LSC, with the setting time increasing as the LS dosage rises. The specific setting times for all tested groups were 30 s, 42 s, 80 s, 95 s, 120 s, and 130 s, none of which exceeded 3 min. Such short setting times are advantageous for adapting to the complex lunar surface environment. The reason for this phenomenon is that in the L-MPC-LSC system, LS particles adsorb both borax and phosphate ions. Compared to borax, LS particles exhibit a more pronounced adsorption effect on phosphate ions, resulting in a reduction of phosphate ions available to react with MgO, thereby delaying the setting time. As the LS dosage increases to replace MgO content, LS becomes the dominant component in the L-MPC-LSC system, while the amounts of MgO powder and phosphate salts decrease. This reduction slows the reaction rate of L-MPC-LSC, thereby extending the material setting time. Additionally, due to the highly similar chemical composition between LS and fly ash, combined with the fact that LS particles, characterized by their irregular polyhedral morphology, possess a large specific surface area, they likely exhibit strong water absorption capacity. Under a constant water-binder ratio, this reduces the effective water-binder ratio, leading to insufficient moisture in the L-MPC-LSC system. The resulting moisture deficiency slows the reaction rate and further prolongs the setting time.

3.2. Mechanical properties

3.2.1. Effect of LS content

Fig. 8 illustrates the effect of different LS dosages on the mechanical properties of L-MPC-LSC under air-curing conditions. As shown, with increasing curing age, the compressive and flexural strengths of L-MPC-LSC specimens with varying LS dosages exhibited

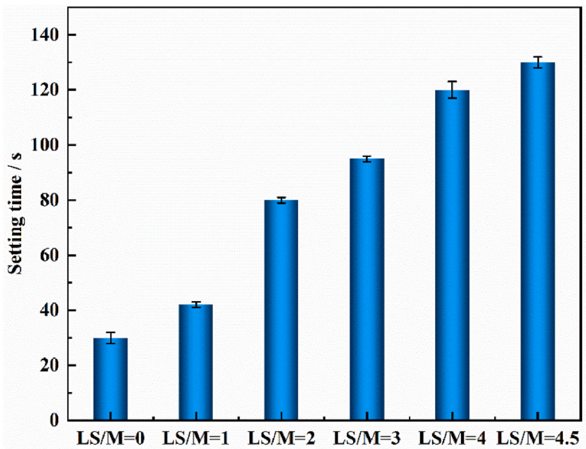

Fig. 7. Effect of LS dosage on the setting time of L-MPC-LSC.

the same growth trend. Both compressive and flexural strengths initially increased and then decreased as the LS dosage rose. As shown in Fig. 8(a), taking the 3-h curing age as an example, when the LS dosage increased from 0 % to 60 %, the compressive strengths of the specimens in the LS/M = 0, LS/M = 1, LS/M = 2, and LS/M = 3 groups were 3.76 MPa, 10.58 MPa, 6.72 MPa, and 5.1 MPa, respectively. Compared to the control group (LS/M = 0), the compressive strengths of the other groups increased by 180 %, 78.7 %, and 35.6 %, respectively. These results indicate that an appropriate LS dosage enhances the mechanical properties of L-MPC-LSC. However, when the LS dosage increased from 60 % to 70 %, the compressive and flexural strengths of L-MPC-LSC at all curing ages exhibited a declining trend compared to the control group, indicating that the high-proportion incorporation of LS adversely affected the mechanical performance of L-MPC-LSC.

To further quantify the strength contribution rate of LS to L-MPC-LSC, this study referenced the method for evaluating the strength contribution rate of the pozzolanic effect proposed by Pu [53]. The relative contribution of LS to L-MPC-LSC strength was initially calculated. The computational formula is given by Equation (6):

$$R = \frac{R_C}{q} \tag{6}$$

Where R is the relative strength contribution (MPa), R_C represents the measured strength (MPa), and q denotes the mass percentage of LS in the L-MPC-LSC system. When calculating the relative contribution of LS-free L-MPC-LSC, q is taken as 100 %.

Subsequently, the strength contribution rate of LS to L-MPC-LSC is calculated using Equations (7) and (8):

$$R_{LS} = R_{L-MPC-LSC} - R_{MPC} \tag{7}$$

$$P = \frac{R_{LS}}{R_{L-MPC-LSC}} \tag{8}$$

Where R_{LS} denotes the specific strength (MPa), $R_{L\text{-}MPC\text{-}LSC}$ represents the relative strength contribution of L-MPC-LSC (MPa), R_{MPC} is the relative strength contribution of LS-free L-MPC-LSC (MPa), and P designates the strength contribution rate of LS to L-MPC-LSC (%).

As shown in Fig. 9, the strength contribution rates of LS in each group to the L-MPC-LSC material at different curing ages were calculated accordingly; when the LS dosage does not exceed 60 % of the total system (LS/M $\leq$ 3), the compressive strength contribution rate of LS to L-MPC-LSC remains stable within the range of 67.9 %–77.3 % across all curing ages, while the flexural strength contribution rate stabilizes between 48.2 % and 66.7 %. This demonstrates that LS exhibits a significant reinforcing effect in the L-MPC-LSC system. However, when the LS dosage exceeds 60 % of the system (LS/M $\geq$ 3), the strength of L-MPC-LSC significantly decreases, and the strength contribution rate of LS even becomes negative. This indicates that excessive LS incorporation adversely affects the mechanical properties of the L-MPC-LSC material.

Relevant studies [3] indicate that the minimum compressive strength of concrete structures ranges between 25 and 30 MPa. Since the gravitational force on the Moon is approximately 1/6 of that on Earth, it can be assumed that any structural strength requirements for concrete on the Moon would also be reduced to 1/6 of terrestrial standards. Consequently, an acceptable compressive strength range for lunar applications is 4–6 MPa. From the compressive strength test results of L-MPC-LSC shown in Fig. 8, when LS/M $\leq$ 3, the 3-h compressive strength of the printed specimens exceeds 4 MPa, meeting the strength requirements for lunar environments. In this case, lunar soil constitutes up to 60 % of the L-MPC-LSC system. Based on the experimental results above, the maximum LS dosage group under air-curing conditions is the LS/M = 3 group, and the compressive strength of L-MPC-LSC falls within the 4–6 MPa range, essentially satisfying the strength demands of the lunar surface environment.

3.2.2. *Effect of simulated lunar vacuum environments*

In the aforementioned study, we analyzed the influence of LS dosage on the strength of L-MPC-LSC under air curing conditions.

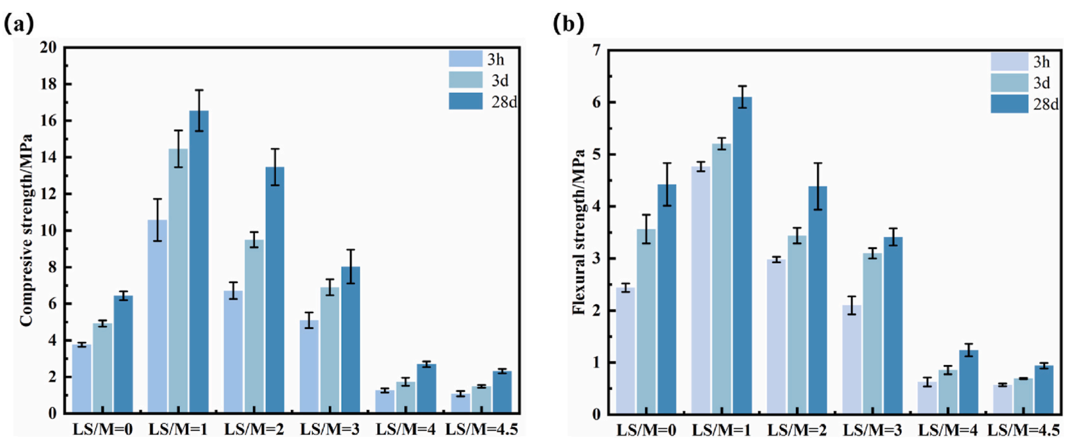

Fig. 8. Effect of LS dosage on compressive strength (a) and flexural strength (b) of L-MPC-LSC

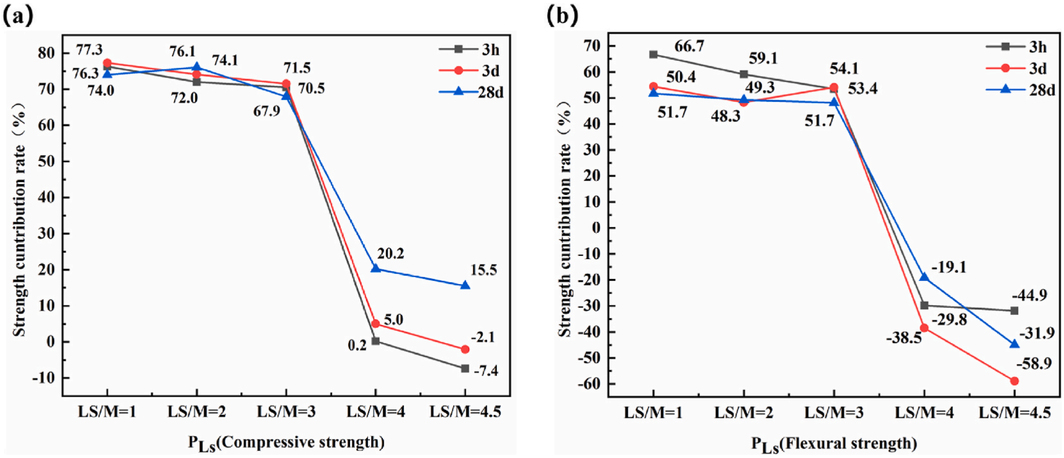

Fig. 9. (a) Compressive strength contribution rate and (b) flexural strength contribution rate of LS in the L-MPC-LSC system.

However, considering the unique high-vacuum environment of the lunar surface, and given that studies have shown vacuum curing of cementitious materials prepared under standard temperature and pressure can limit strength development [54], this paper focuses on LS dosages ≤60 % in simulated vacuum environments.

Fig. 10 shows the strength variations of LS/M = 1, LS/M = 2, and LS/M = 3 groups under simulated lunar vacuum environments. As clearly observed from the graph, the strength of L-MPC-LSC still exhibits a growth trend with prolonged curing time. This phenomenon indicates that under vacuum conditions, the hydration process of L-MPC-LSC has not halted but remains ongoing, although its strength development may be constrained to some extent. As shown in Fig. 11, under vacuum curing conditions, all experimental groups exhibited reduced strength at all curing ages compared with air curing. The flexural strength losses ranged between 25 % and 52 %, while compressive strength losses fell within 19 %–41 %. These results demonstrate that vacuum curing significantly affects the mechanical properties of L-MPC-LSC. As shown in Fig. 10(a), taking the 3-h curing age as an example, the compressive strengths of the specimens under vacuum curing are 7.37 MPa, 5.43 MPa, and 3.10 MPa for the respective groups, indicating that 3D-printed specimens meet the strength requirements for lunar environments when LS/M ≤ 2. At the 3-day curing age, the compressive strengths of the groups increase to 9.48 MPa, 6.6 MPa, and 4.1 MPa. For the LS/M = 3 group, although the flexural strength of specimens under vacuum curing decreased by 51.3 % and compressive strength reduced by 40.6 % compared to air curing, the L-MPC-LSC specimens with LS/M = 3 still maintained a compressive strength of 4.1 MPa at 3 days. This strength level meets the lunar environment's requirements. Based on the experimental results above, it can be concluded that the L-MPC-LSC material prepared with the LS/M = 3 group retains the required mechanical properties even when exposed to a vacuum environment after undergoing a short curing period.

3.3. Microstructural characteristics

Fig. 12 shows the influence of different LS dosages on the XRD patterns of L-MPC-LSC specimens under air curing. The analysis results indicate that the diffraction peaks of the L-MPC-LSC specimens are mainly composed of K-Struvite ($MgKPO_4·6H_2O$), Dittmarite

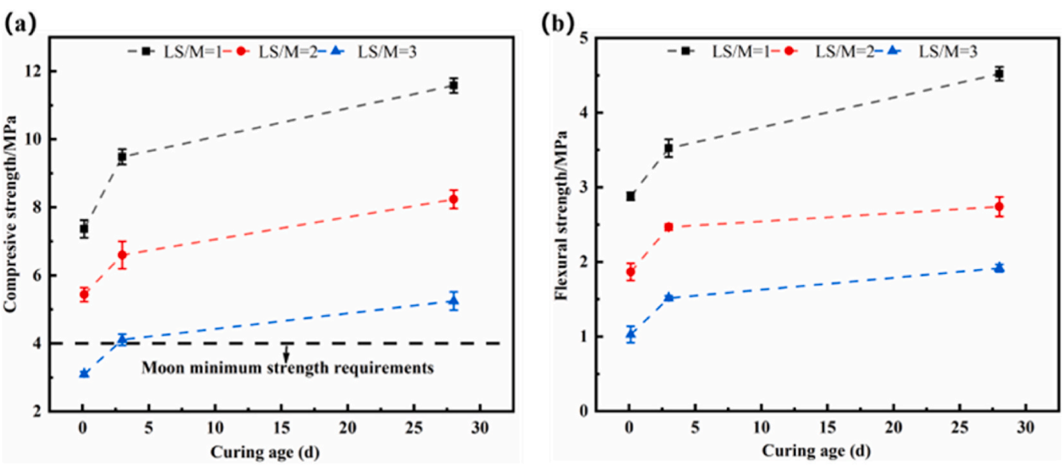

Fig. 10. (a) Compressive strength and (b) Flexural strength of L-MPC-LSC in a vacuum curing.

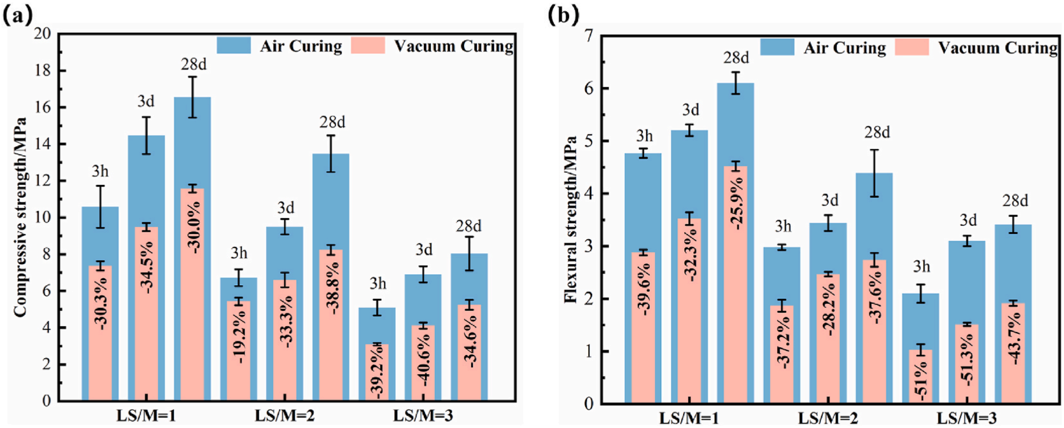

Fig. 11. (a) Compressive strength comparison and (b) flexural strength comparison of L-MPC-LSC under air and vacuum curing conditions.

($MgKPO_4 \cdot H_2O$), $MgCO_3$, SiO_2, and unreacted MgO. Compared with the control group (LS/M = 0), the diffraction peak intensity of Dittmarite in LS-incorporated specimens decreases and gradually diminishes with increasing LS dosage. In contrast, the diffraction peak intensity of K-Struvite exhibits an initial increase followed by a decrease, indicating that the incorporation of LS affects the crystallinity and quantity of K-Struvite. This phenomenon is attributed to the high reactivity of light-burned MgO, which triggers a vigorous and exothermic reaction. Such intense reactivity not only inhibits the formation of additional hydration products [55] but also induces dehydration and decomposition of the hydration products [56], thereby resulting in low-intensity Dittmarite diffraction peaks in the control group. However, when a certain amount of LS is incorporated, part of the light-burned MgO is replaced by LS, reducing the total reactant quantity and slowing the reaction rate. This allows the hydration reaction to proceed more completely, which leads to the formation of K-Struvite crystals with higher strength than Dittmarite in the L-MPC-LSC system, thereby improving the strength of the specimens. The quantitative analysis results from Topas 5.0 software (Table 3) further confirm this observation: the K-Struvite content in the LS/M = 1 group increased to 45.80 % compared to the control group, while the Dittmarite content significantly decreased from 42.43 % to 7.67 %. This compositional shift explains the superior mechanical properties of the LS/M = 1 group over the LS/M = 0 group. When the LS dosage increased from 60 % to 70 %, both the compressive and flexural strengths of 3D printing L-MPC-LSC at all curing ages showed a decreasing trend compared to the control group. The reason for this phenomenon is as follows: As the replacement of MgO and phosphate by LS increases, the proportions of MgO and phosphate in the system progressively decrease, allowing LS to dominate the composition. This reduces the formation of hydration products, thereby further diminishing the strength of the L-MPC-LSC. Although the micro-aggregate effect of LS can enhance the strength of MPC-LSC, it is insufficient to compensate for the strength reduction caused by the decreased formation of hydration products. This phenomenon is consistent with findings in studies of carbon nanotube-reinforced cement-based composites, where exceeding the critical filler dosage is observed to impair mechanical properties due to agglomeration effects [57,58]. The quantitative analysis results (Table 3) reveal that when comparing the LS/M = 3 and LS/M = 4.5 groups, the relative content of K-Struvite in the system decreased from 25.08 % to 18.08 %. This trend

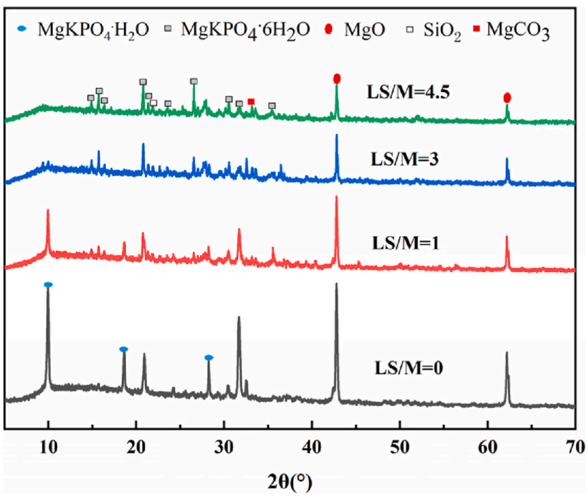

Fig. 12. XRD patterns of L-MPC-LSC with different LS dosages after 28 days of curing
Note: K-Struvite (PDF#15–0762), Dittmarite (PDF#36–1491), MgO (PDF#30–0794),SiO_2 (PDF#46–1045), $MgCO_3$ (PDF#08–0479).

Table 3
Quantitative analysis results (wt. %) of the phase composition of L-MPC-LSC.

	$MgCO_3$	MgO	$MgKPO_4{\cdot}6H_2O$	$MgKPO_4{\cdot}H_2O$	KH_2PO_4	B	SiO_2	LS
LS/M = 0	6.40	35.37	11.29	42.43	–	–	4.51	–
LS/M = 1	2.93	8.20	45.80	7.67	–	–	2.07	33.33
LS/M = 3	2.32	11.32	25.08	–	–	–	1.28	60.00
LS/M = 4.5	0.89	10.12	18.18	–	–	–	1.54	69.20

Note: LS content denotes its mass ratio within the L-MPC-LSC system; B represents $Na_2B_4O_7{\cdot}10H_2O$.

reasonably explains the progressive decline in specimen strength caused by the gradual increase in LS dosage.

Fig. 13 compares the microstructures of the LS/M = 0 and LS/M = 3 specimens at a curing age of 28 days. As shown in the images, the LS-free specimen (LS/M = 0) displays a large number of plate-like Dittmarite crystals and a small amount of K-Struvite crystals under the same magnification. These crystals are interlaced, resulting in a loose internal structure with pronounced gaps between hydration products, which is one of the reasons for its lower strength. In contrast, the LS-incorporated specimen (LS/M = 3) exhibits a more compact microstructure. Plate-like Dittmarite crystals and blocky K-Struvite crystals are nearly absent, while LS particles effectively fill the gaps and become gradually encapsulated by the surrounding cementitious matrix, demonstrating good chemical compatibility. This indicates that the addition of LS significantly enhances the interfacial bonding within the L-MPC-LSC structure and improves its overall densification, thereby increasing the strength of L-MPC-LSC.

Fig. 14 presents the XRD patterns of the LS/M = 3 group L-MPC-LSC specimens under vacuum curing conditions. It can be observed that the crystalline composition remains consistent with that of air-cured L-MPC-LSC specimens, showing no significant changes. The primary crystalline phases include $MgKPO_4{\cdot}6H_2O$ (K-Struvite), $MgCO_3$, SiO_2, and unreacted MgO. Quantitative analysis using Topas 5.0 software (Table 4) revealed that the content of K-Struvite crystals decreased from 25.08 % to 19.56 %. This reduction is attributed to water loss from the L-MPC-LSC system under vacuum conditions, resulting in a water-deficient state that hinders the hydration reaction between MgO and phosphate, thereby reducing the formation of hydration products. Given that K-Struvite is the primary contributor to the strength of MPC [34,59], this variation reasonably explains the strength reduction of L-MPC-LSC specimens under vacuum curing compared to air curing. Furthermore, as shown in Fig. 15, a comparative analysis of the microstructures of the LS/M = 3 group specimens at 28 days under different curing environments reveals that rapid water loss in the vacuum environment leads to incomplete hydration, resulting in reduced cementitious material formation. This contributes to a loose overall structure and a significant increase in porosity, which is another key factor explaining the strength reduction of specimens under vacuum curing compared to air curing.

3.4. Evaluation of 3D printing performance and applications

3.4.1. Extrudability and buildability

Following an in-depth analysis of the effects of varying LS dosages on the setting time and mechanical properties of L-MPC-LSC, it becomes evident that the LS/M = 3 formulation demonstrates optimal comprehensive performance. This ratio achieves the maximum utilization of in-situ resources while ensuring compliance with strength requirements, making it the most balanced candidate for practical applications. Therefore, this formulation was selected for further evaluation of its 3D printing performance for L-MPC-LSC. As shown in Fig. 16(a), the printed filaments from the LS/M = 3 group exhibit continuous and uniform deposition. Measurements indicate

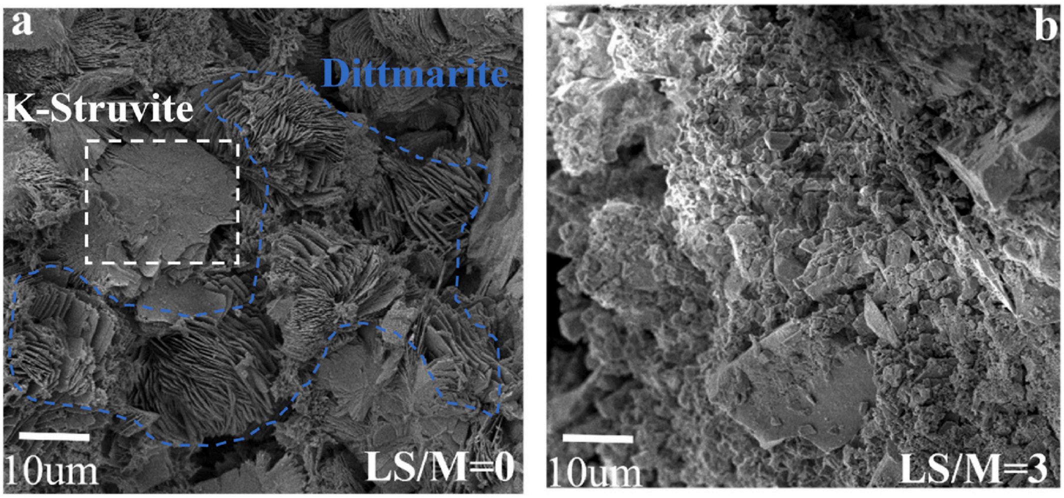

Fig. 13. Micro-morphology of L-MPC-LSC after 28 days of curing observed by SEM.

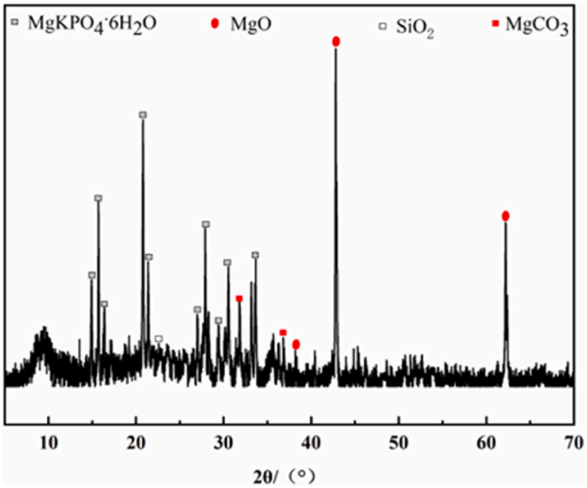

Fig. 14. XRD patterns of L-MPC-LSC samples after 28 days of curing under vacuum
Note: K-Struvite (PDF#15–0762), MgO (PDF#30–0794),SiO_2 (PDF#46–1045), $MgCO_3$ (PDF#08–0479).

Table 4
Quantitative analysis results (wt. %) of phase composition for L-MPC-LSC (LS/M = 3) under different curing environments.

	$MgCO_3$	MgO	$MgKPO_4{\cdot}6H_2O$	KH_2PO_4	B	SiO_2	LS
Air curing	2.32	11.32	25.08	–	–	1.28	60.00
Vacuum curing	3.32	14.16	19.56	–	–	2.96	60.00

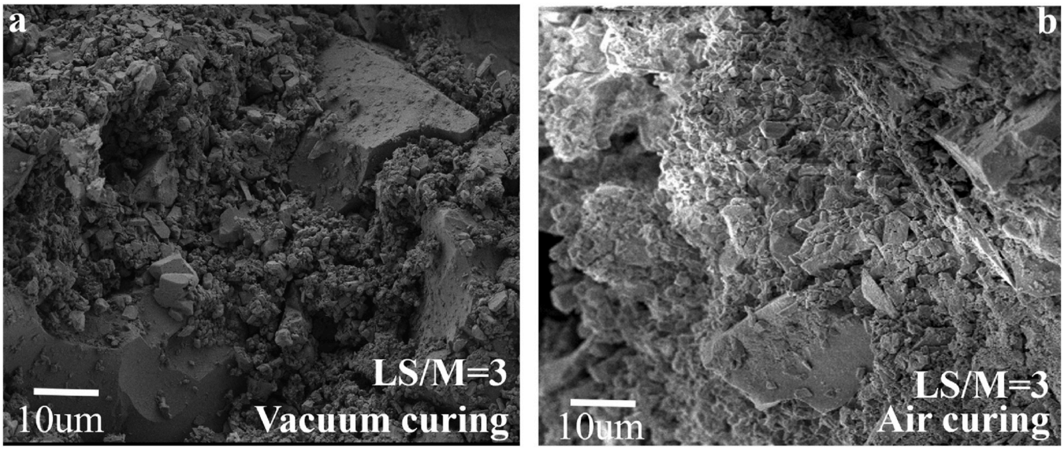

Fig. 15. SEM micro-morphology of L-MPC-LSC after 28 days of curing under different environments.

an average filament width of 19.6 mm and thickness of 9.95 mm. The calculated width uniformity (coefficient of variation) is 2.04 %, while thickness uniformity reaches 0.48 %, both meeting the predefined technical specifications for lunar construction-grade printable materials. Furthermore, no instances of filament breakage or discontinuity were observed throughout the printing process, nor did issues such as drying-induced defects or overly fluid behavior occur. This demonstrates excellent homogeneity and process stability. The observed performance is primarily attributed to the delayed setting time of L-MPC-LSC caused by LS incorporation, which enhances the material's flowability and thereby ensures superior printability.

Fig. 16(b) shows the test results of the 3D-printed hollow cylindrical component. Measurements indicate that the printed component has an average height of 203.55 mm and an average diameter of 153.1 mm. Compared to the designed dimensions, the height deviates by only 0.22 %, while the average diameter deviation is 2.07 %, resulting in an overall average dimensional deviation of 1.14 % for both diameter and height. These results demonstrate that the L-MPC-LSC material exhibits excellent buildability. Additionally, no significant collapse or cracking was observed in the printed components. This is primarily attributed to the rapid hydration rate and high early-age strength of the L-MPC-LSC material. During printing, each deposited layer rapidly sets and hardens

Fig. 16. (a) Extrudability and (b) printing constructability of structural components of L-MPC-LSC material.

after extrusion, developing sufficient strength to support the subsequent layer. This hierarchical curing mechanism ensures process stability and preserves the structural integrity of the printed components.

3.4.2. 3D printing application of lunar lava tube habitat dome structural models

Based on the evaluation results of the 3D printing performance of the L-MPC-LSC material, its demonstrated excellent printability provides robust support for the fabrication of lunar lava tube habitat domes. To further validate the feasibility of printing the dome structure shown in Fig. 17(a), this study developed a tailored design for the lunar lava tube habitat dome model by integrating the operational characteristics of 3D printing equipment with the unique environmental constraints of the Moon.

In the preliminary phase of this study, we systematically compared hemispherical structures, arch structures, and cylindrical configurations for lunar applications. Through comprehensive evaluation, the arch structure was identified as the optimal choice for lunar environmental compatibility. Building upon this foundation, we propose an innovative dome design that synergistically integrates the load-bearing efficiency of arch structures with the geometric resilience of honeycomb cellular configurations. This integration has yielded a novel arc-curved hexagonal dome structural model, as shown in Fig. 17(b), specifically engineered for lunar lava tube habitat construction. The dome structure employs internal arch configurations to enhance load-bearing capacity. Its unique petal-shaped edge design facilitates the integration of multi-walled carbon nanotubes (MWCNTs) within each petal interlayer,

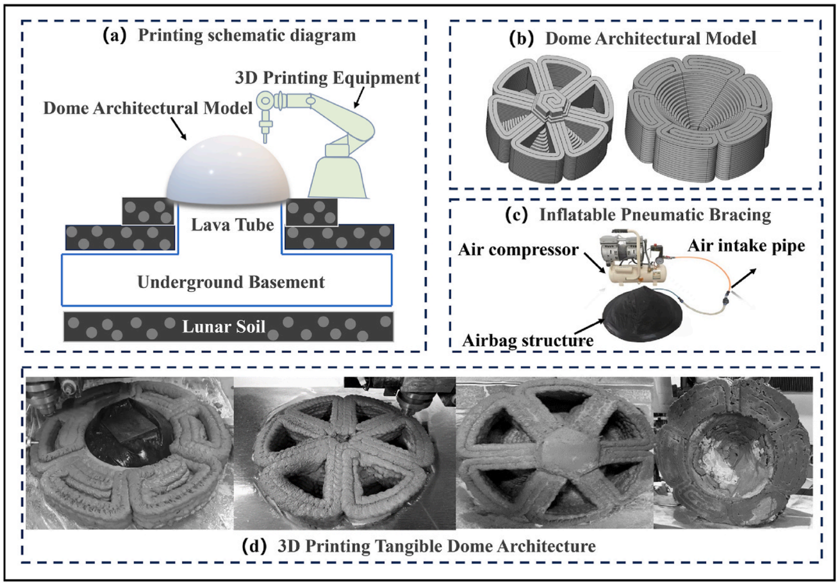

Fig. 17. Flowchart of dome model design and 3D printing.

reinforcing the mechanical properties of the cement-based composite. This MWCNT-enhanced composite exhibits lightweight yet high-strength characteristics, exceptional dynamic load adaptability, and high stability in extreme environments, collectively improving the dome's seismic resilience under dynamic loading while extending its service life in lunar base conditions [60]. The external hexagonal configuration draws inspiration from honeycomb structures, which are extensively utilized in aerospace and civil engineering applications due to their high specific strength and specific stiffness. Furthermore, constructing such a dome atop lunar lava tubes presents significant technical challenges due to the enormous scale of the lava tube openings. To address this extreme environmental constraint and enable successful dome printing, the development of a specialized support structure becomes imperative. To achieve this objective, this study designed an inflatable airbag for the aforementioned dome model, as shown in Fig. 17(c). This airbag can be dynamically inflated or deflated during the 3D printing process to adjust the dome's geometry and structural configuration in real time. Such an innovative design not only meets the predefined quality and geometric standards for the dome structure but also addresses the cost challenges of traditional support systems and overcomes the self-weight bearing limitations inherent to large-scale lunar constructions. By enabling adaptive shape modulation, the system ensures structural stability throughout the printing cycle while minimizing material waste and logistical complexity.

The dimensional parameters of the dome structure model are as follows: the radius of the outer ring at the base layer is 333 mm, with an inner diameter of 232.5 mm; the total height is 252.0 mm, divided into 21 layers. During the layer-by-layer printing process, the inner radius of each layer decreases by 12.5 mm compared to the previous layer. The arched structure inside the model has a height of 168 mm, while the hexagonal structure at the top has a height of 60 mm. The design transitions from the arched structure to the hexagonal form at Layer 16, integrating these geometries to effectively distribute stress and support loads.

As shown in Fig. 17(d), using the hybrid-mixing-stirring-extrusion integrated rapid-setting concrete 3D printing equipment illustrated in Fig. 3, combined with the inflatable dome support structure depicted in Fig. 17(c), the dome structure model was successfully completed after 120-min continuous 3D printing. The entire printing process remained stable with no material delamination or structural collapse, demonstrating the application potential of the L-MPC-LSC material in continuous printing technology through the integrated hybrid-mixing-stirring-extrusion 3D printing system.

Based on the unique geometric morphology of the above-mentioned arc-curved hexagonal dome, this study proposes an innovative construction strategy for future lunar base dome construction. As shown in Fig. 18(a), the strategy integrates 3D printing technology with lunar rovers to enable automated and remote construction operations. Through this approach, construction personnel can remotely control multiple 3D printing devices via computers within the space capsule to perform synchronized printing operations. Each device is responsible for printing a petal-shaped unit of the dome structure, enabling collaborative multi-robot synchronous printing within a small robotic team. To further investigate the feasibility of the construction strategy for lunar base dome fabrication through collaborative multi-robot synchronous printing as shown in Fig. 18(a), this study divides the integrated dome structure shown in Fig. 17(a) into six petal-shaped modular units for separate printing. The printing paths and results are shown in Fig. 18(b), where the petal-shaped modular units were successfully printed with precise geometric dimensions. Each unit exhibited a smooth and flat surface alongside a uniform and dense internal structure, fully meeting the design requirements. Fig. 18(c) shows the integrated structure assembled from six individually printed petal-shaped units. The results indicate that the overall morphology of the dome aligns closely with the design intent, with only minor gaps observed at the joints between adjacent units. These slight discrepancies may stem from alignment precision between units and cumulative errors during the printing process. Despite these minor deviations, the structural stability and integrity of the dome remain unaffected, validating the feasibility and effectiveness of collaborative multi-robot synchronous printing in constructing large-scale complex architectures with precision. Notably, the printing time for each petal-shaped

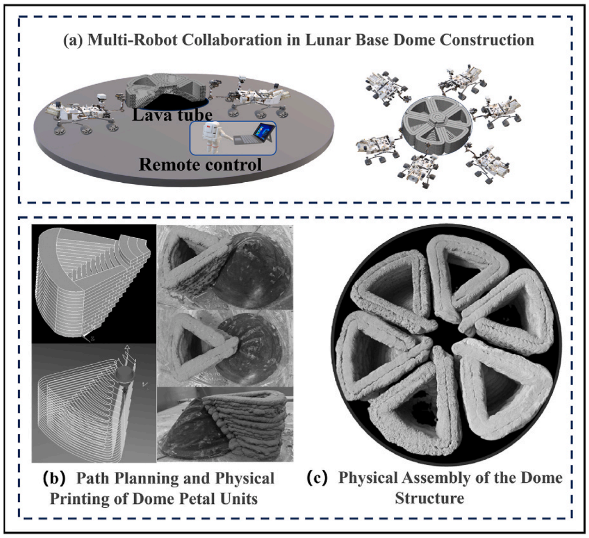

Fig. 18. Schematic diagram of multi-robot collaborative 3D printing and the fabricated structure.

unit was strictly controlled within 20 min. This rapid printing process not only demonstrates the robotic team's efficient collaboration but also highlights the significant advantages of this strategy for future lunar lava tube base construction.

4. Conclusions

Based on the concept of lunar soil resource utilization, this study successfully developed a novel L-MPC-LSC material. Through systematic investigation of the effects of lunar soil content on the mechanical properties and hydration product composition of L-MPC-LSC under both standard atmospheric conditions and vacuum curing conditions, we determined the optimal mix proportion suitable for 3D printing in lunar vacuum environments. Utilizing a self-developed 3D printing system with integrated mixing, stirring, and extrusion functions, we successfully fabricated a prototype of lunar lava tube habitat dome structure. The main conclusions are as follows.

(1) The addition of LS effectively improved the setting time of L-MPC-LSC and significantly enhanced its mechanical properties when incorporated in appropriate amounts. Nevertheless, when the LS content surpassed 60 % (LS/M $\geq$ 3), the material strength diminished considerably, indicating that excessive incorporation of LS is detrimental to the mechanical properties of the material.

(2) Quantitative XRD analysis revealed that the Dittmarite crystal content linearly decreased from 42.43 % to 0 % with increasing LS incorporation, while the K-Struvite crystal proportion exhibited an initial increase (peaking at 45.80 %) followed by a subsequent decrease. The synergistic phase transformations between these two crystalline phases were responsible for the observed strength variation of L-MPC-LSC, characterized by an initial enhancement followed by degradation. SEM characterization demonstrated that LS particles acted as micro-aggregates to fill voids and optimize particle distribution, thereby forming a compact structure that enhanced the mechanical strength.

(3) The vacuum environment induced water evaporation from the L-MPC-LSC system, resulting in suppressed hydration reactions between MgO and phosphate salts. This inhibition mechanism led to reduced K-Struvite crystal formation and subsequent structural deterioration, ultimately causing strength reductions ranging from 19 % to 52 %. Notably, at 60 % LS incorporation, the material maintained a 3-day compressive strength of 4.1 MPa, demonstrating an optimal balance between mechanical performance and lunar soil resource utilization.

(4) The L-MPC-LSC material demonstrated superior printability with coefficients of variation of 2.04 % for printed filament width and 0.48 % for thickness. Circular structural components exhibited a mean overall deviation of 1.14 % in both diameter and height dimensions, confirming precise dimensional control during the extrusion process.

(5) Through the design and implementation of multi-unit printing experiments featuring arc-curved hexagonal dome structures, the application potential of L-MPC-LSC materials in extraterrestrial construction was successfully validated. The integrated system combining continuous 3D printing technology with robotic collaborative construction techniques showed promising compatibility for lunar lava tube dome engineering applications.

CRediT authorship contribution statement

Libo Lyu: Writing – original draft, Visualization, Investigation, Data curation. **Weihong Li:** Writing – review & editing, Supervision, Project administration, Conceptualization. **Yongjie Deng:** Visualization, Methodology, Investigation. **Qiuchun Yu:** Visualization, Formal analysis. **Haiyan Ma:** Supervision, Conceptualization. **Hongfa Yu:** Writing – review & editing, Resources, Formal analysis, Conceptualization. **Lingyu Li:** Visualization, Methodology. **Haosong Xuan:** Investigation. **Honglei Zhang:** Investigation. **Mingyang Lu:** Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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